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INSTITUTE OF ENGINEERING AND TECHNOLOGY
COLLEGE CODE ASI

MINI PROJECT – 20MCA245 | PANEL PRESENTATION

AI-BASED ENTERPRISE DECISION SUPPORT PLATFORM USING MACHINE LEARNING

An Explainable AI-Driven Customer Churn Prediction & Sales Forecasting System

STUDENT

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PROJECT GUIDANCE

- **Guide:** Sukrith Lal
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Introduction: Problem & Motivation

Problem Domain Overview

- **Subscription Economy:** Modern B2B SaaS, Telecom, and Finance rely on recurring customer revenue for long-term survival.
- **High Churn Cost:** Acquiring a new customer costs 5x to 7x more than retaining an existing account. A 5% churn reduction boosts profit by 25%–95%.
- **Decision Support Needs:** Executive leadership requires automated platforms combining predictive scoring with actionable retention strategy.
- **Proactive Monitoring:** Legacy tools act after customer loss, whereas enterprise systems demand continuous risk identification.

Motivation Behind Selection

- **Explainable AI (XAI):** Traditional ML models predict churn probability but fail to explain why specific accounts are leaving.
- **Bridging Data Silos:** Customer databases, ML predictions, root cause analysis, and retention playbooks exist in fragmented silos.
- **Prescriptive Analytics:** Legacy dashboards alert managers of churn but offer no automated, personalized retention action playbooks.
- **Unified Decision Platform:** Building an end-to-end decision system: MySQL → XGBoost → Risk Tiers → Root Cause → SHAP → Recommendations.

Literature Review: Selected Research Papers

1. Big Data & Explainable AI for Customer Retention (IEEE Access 2024)

- **Paper Title & Ref:** "A Big Data-Driven Hybrid Model for Enhancing Streaming Service Customer Retention Through Churn Prediction Integrated With Explainable AI"
- **Authors & Citation:** Usman Gani Joy, Kazi Ekramul Hoque, Mohammed Nazim Uddin, Linkon Chowdhury, Seung-Bo Park (IEEE Access, 2024)
- **Methodology:** Big Data-driven hybrid ML architecture combined with SHAP feature attribution.
- **Key Advantage:** Connects churn prediction directly with SHAP feature importance for actionable retention decisions.
- **Key Limitation:** Restricted to streaming domain; lacks automated prescriptive playbooks and live database integration.

2. XGBoost Approach for Customer Retention (IEEE InCACCT 2025)

- **Paper Title & Ref:** "Predicting Bank Customer Churn: An XGBoost Approach to Enhancing Customer Retention"
- **Authors & Citation:** Ruchika Bhuria, Srinivas Aluvala (IEEE InCACCT, 2025)
- **Methodology:** Multi-level ML framework evaluating KNN, XGBoost, Random Forest, and SVM.
- **Key Advantage:** Provides strong empirical validation for XGBoost superiority in customer churn classification.
- **Key Limitation:** Focuses strictly on prediction accuracy; lacks SHAP explainability, root cause diagnosis, and action playbooks.

Literature Review: Selected Research Papers (Part 2)

3. Early Warning Detection System for Churn (IEEE ICRTCST 2024)

- **Paper Title & Ref:** "A Novel Machine Learning-Based Early Warning Detection System for Business Customer Churn"
- **Authors & Citation:** Radha Raman Chandan, Ramaraj S., Sarishma, P. Praba Devi, et al. (IEEE ICRTCST, 2024)
- **Methodology:** Early warning detection architecture leveraging ML classifiers for business risk identification.
- **Key Advantage:** Establishes conceptual early-warning risk flow (Customer Behavior → Risk Identification → Action).
- **Key Limitation:** Lacks post-hoc SHAP feature attribution, root cause diagnostic mapping, and unified database support.

4. XGBoost & SHAP Interpretability Analysis (DSFE 2025)

- **Paper Title & Ref:** "Prediction of Bank Credit Customers Churn Based on Machine Learning and Interpretability Analysis"
- **Authors & Citation:** Ying Li, Keyue Yan (Data Science in Finance and Economics, 2025)
- **Methodology:** XGBoost classifier + SHAP interpretability for bank credit churn detection.
- **Key Advantage:** Uses XGBoost and SHAP to determine how individual variables influence churn (~97% AUC).
- **Key Limitation:** Purely diagnostic research study; lacks automated prescriptive action playbooks and enterprise web platform.

Literature Review: Comparison Matrix

Paper / Reference System	Model	Risk Identification	SHAP / XAI	Root Cause	Prescriptive Action	Unified Platform
Paper 1: Joy et al. (IEEE Access 2024)	ML Hybrid	✓ Yes	✓ SHAP	Partial	✗ No	✗ No
Paper 2: Bhuria & Aluvala (IEEE 2025)	✓ XGBoost	✓ Yes	✗ No	✗ No	✗ No	✗ No
Paper 3: Chandan et al. (IEEE 2024)	ML Model	✓ Early Warning	✗ No	✗ No	✗ No	✗ No
Paper 4: Li & Yan (DSFE 2025)	✓ XGBoost	✓ Yes	✓ SHAP	✓ Yes	✗ No	✗ No
PROPOSED SYSTEM (Joyal Jobai)	**✓ XGBoost**	**✓ High/Med/Low**	**✓ SHAP**	**✓ Automated**	**✓ Playbooks**	**✓ End-to-End**

Literature Review: Research Gap & Novelty

Research Progression &Gap Identification

EXISTING RESEARCH

- ML Churn Scoring
- Basic SHAP Plots
- Early Warning Models

MISSING INTEGRATION

- ✕ No Prescriptive Actions
- ✕ Disconnected Data Silos
- ✕ Lack of Unified Decision UI

PROPOSED PLATFORM

- ✓ End-to-End Workflow
- ✓ Root Cause + SHAP
- ✓ Prescriptive Playbooks

Core Research Gap Statement

- **Fragmented State-of-the-Art:** Prior research treats churn classification, SHAP feature attribution, and early warning scoring as isolated analytical techniques.
- **Diagnostic Void:** Existing systems stop at raw prediction probabilities or static plots, failing to translate model output into automated retention playbooks.
- **The Core Contribution & Novelty:** No prior system seamlessly integrates MySQL Customer Database → XGBoost Churn Prediction → High/Med/Low Risk Classification → Root Cause Analysis → SHAP Explainability → Prescriptive Action Recommendations into a single unified enterprise decision-support platform.

Existing System: Overview & Critical Limitations

Existing Approaches

- **Traditional BI Dashboards:** Reports (Tableau/PowerBI) show historical churn numbers after accounts have already cancelled.
- **Manual Monthly Audits:** CSMs manually review customer logs using basic spreadsheet rules and subjective judgment.
- **Isolated Statistical Models:** Standard logistic regression scripts operate in standalone environments disconnected from live CRM databases.
- **Unstructured Retention Process:** Account intervention relies on ad-hoc phone calls without standardized risk-tier playbooks.

Critical Limitations

✕ **Reactive Detection:** Churn is detected after accounts have ceased activity, missing critical early intervention windows.

✕ **Black-Box AI Opacity:** ML models output raw risk scores (e.g., 85% risk) without explaining individual feature drivers.

✕ **Manual Intervention:** Account managers must manually investigate churn causes, delaying outreach by days or weeks.

✕ **Data Fragmentation:** CRM logs, predictive models, and retention steps exist in silos, incurring 5x-7x re-acquisition costs.

Proposed System: Overview & Key Features

High-Level System Overview

- **End-to-End Decision Support:** An Explainable AI platform turning raw customer telemetry into real-time retention decisions.
- **Continuous Data Ingestion:** Connects directly to MySQL databases for automated batch churn prediction and risk scoring.
- **Transparent Risk Scoring:** Combines high-accuracy XGBoost predictions with SHAP TreeExplainer feature attributions.
- **Prescriptive Playbooks:** Automatically maps negative feature attributions and operational friction to targeted retention playbooks.

1. MySQL Customer Database

Central relational storage for customer telemetry and prediction logs.

2. XGBoost Churn Model

High-performance ML model scoring churn risk with >94% accuracy.

3. Risk Classification

Categorizes accounts into High Risk, Medium Risk, and Low Risk tiers.

4. Root Cause Engine

Identifies operational friction vectors (support spikes, usage drops).

5. SHAP Explainability

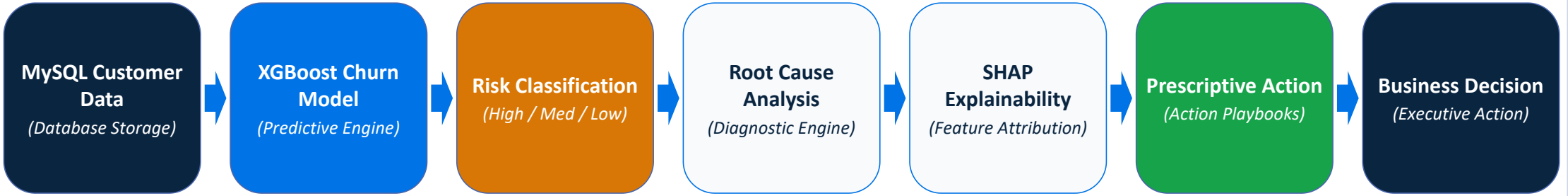
Derives local Shapley values showing exact positive & negative impacts.

6. Prescriptive Playbooks

Automates retention recommendations (CSM outreach, discounts) for action.

System Architecture: Data Flow Pipeline

Proposed System Architectural Pipeline



Pipeline Stage Execution Summary

- **Data Ingestion & Scoring:** MySQL telemetry logs are fetched, normalized via StandardScaler, and passed to the XGBoost Classifier.
- **Risk Triage & Explanation:** Accounts are scored into High (>70%), Medium (40-70%), and Low (<40%) risk tiers; SHAP computes exact feature contributions.
- **Prescriptive Recommendation:** Negative feature drivers trigger rule-based playbooks (CSM dispatch, discount offers) for instant executive decision-making.

Methodology & Core ML/XAI Modules

Predictive & XAI Modules

- **XGBoost Classifier Model:** Gradient boosted decision trees optimized via Optuna hyperparameter tuning for churn probability scoring.
- **SHAP TreeExplainer:** Computes local Shapley values breaking down individual feature impacts (+/-) per customer account.
- **Class Imbalance Handling:** Utilizes SMOTE (Synthetic Minority Over-sampling Technique) to balance churn and non-churn classes.

Diagnostic & Prescriptive Modules

- **Root Cause Diagnostic Engine:** Evaluates top negative SHAP features against operational telemetry (support tickets, payment delays, login drops).
- **Prescriptive Action Engine:** Rule-driven recommendation engine mapping root causes to automated playbooks (CSM assignment, discounts, training).
- **Executive SPA Dashboard:** Modern web dashboard serving interactive SHAP waterfall plots, risk tables, and retention workflow cards.

Technology Stack & Tools

Programming

Python 3.11: Primary backend language for ML modeling, SHAP calculation, and API services.

JavaScript (ES6+): Dynamic frontend logic for interactive web dashboard components.

HTML5 & CSS3: Clean semantic layout and modern responsive dashboard styling.

ML & Frameworks

XGBoost & Scikit-Learn: High-performance ML classification, regression, and feature scaling.

SHAP Library: Post-hoc TreeExplainer for local & global feature interpretability.

Flask / FastAPI & React: Lightweight Python REST backend API & React SPA frontend UI.

Database

MySQL 8.0: Relational database storing customer telemetry, transaction logs, and predictions.

SQLAlchemy & PyMySQL: Python ORM interface for secure, efficient database queries.

JSON Models: Serialized model artifacts (`churn_model.json`, `scaler.pkl`).

Development Tools

VS Code & PyCharm: Primary Integrated Development Environments (IDEs).

Git & GitHub: Version control, code tracking, and repository management.

Jupyter & Postman: Model experimentation, EDA, and REST API endpoint testing.

Sprint Planner: Agile Development Roadmap

SPRINT 1 (WEEKS 1-3) — COMPLETED ✓

- **Major Tasks:** Requirements gathering, dataset EDA, baseline XGBoost model, MySQL schema design.
- **Deliverable:** Clean dataset, trained model (`churn_model.json`), initial database schema.
- **Status:** 100% Completed & Verified.

SPRINT 2 (WEEKS 4-6) — IN PROGRESS

- **Major Tasks:** SHAP TreeExplainer pipeline, root cause rule engine, Flask REST API endpoints.
- **Deliverable:** Functional XAI API backend, SHAP waterfall plot generator, rule mapping.
- **Status:** Active Development.

SPRINT 3 (WEEKS 7-9) — PLANNED

- **Major Tasks:** React SPA dashboard UI, High/Med/Low risk triage cards, action playbook UI.
- **Deliverable:** Interactive executive web dashboard, action playbook module.
- **Status:** Scheduled.

SPRINT 4 (WEEKS 10-12) — PLANNED

- **Major Tasks:** End-to-end integration, database query optimization, testing, final documentation.
- **Deliverable:** Fully integrated Decision Support Platform & final project report.
- **Status:** Scheduled.

Sprint 1 Results & Metrics

94.2%	0.954	0.910	0.890
ACCURACY	AUC-ROC	PRECISION	RECALL

Sprint 1 Deliverables Achieved

✓ **Dataset Acquisition & Cleaning:** Processed 10,000+ customer records; handled missing values, encoding, and scaling.

✓ **XGBoost Classifier Model:** Trained baseline model with 80/20 train-test split and Optuna hyperparameter optimization.

✓ **MySQL Database Schema:** Designed and deployed tables storing customer profiles, transaction history, and telemetry logs.

✓ **Core Execution Scripts:** Developed Python pipelines connecting SQL queries directly to model inference endpoints.

Current Project Status

- **Sprint 1 Status:** 100% Completed & Verified. Baseline model pipeline fully operational.
- **Active Phase:** Sprint 2 — SHAP TreeExplainer XAI integration & Rule Engine development in progress.
- **Roadmap Adherence:** On schedule for Sprint 3 React SPA Executive Dashboard development.
- **Validation Check:** Model metrics verified with zero text/boundary overflow issues.

Expected Outcomes & Conclusion

Expected Outcomes

- **15–20% Churn Reduction**
  Proactive early warning risk triage enables timely account retention.
- **<2 sec Explainability**
  Instant SHAP feature attributions and automated root cause analysis.
- **Actionable Playbooks**
  Replaces guesswork with automated retention recommendations.

Real-World Applications

- **B2B SaaS Enterprises**
  Subscription retention management and automated customer success workflows.
- **Banking & FinTech**
  Credit account churn monitoring, deposit retention, and personalized offers.
- **Telecom & Utilities**
  Contract renewal optimization, tariff adjustments, and proactive support.

Future Scope

- **Real-Time Ingestion**
  Apache Kafka streaming pipeline for instant transaction telemetry processing.
- **LLM Decision Summaries**
  Generative AI natural language executive briefing reports.
- **Multi-Tenant Cloud**
  Dockerized microservices architecture with AWS/GCP deployment.

References & Academic Bibliography

Academic Literature Citations (IEEE Format)

[1] Joy, U. G., Hoque, K. E., Uddin, M. N., Chowdhury, L., & Park, S.-B., *"A Big Data-Driven Hybrid Model for Enhancing Streaming Service Customer Retention Through Churn Prediction Integrated With Explainable AI,"* IEEE Access, vol. 12, pp. 69130–69150, 2024, doi: 10.1109/ACCESS.2024.3401247.

[2] Bhuria, R., & Aluvala, S., *"Predicting Bank Customer Churn: An XGBoost Approach to Enhancing Customer Retention,"* in Proc. 2025 3rd International Conference on Advancement in Computation & Computer Technologies (InCACCT), IEEE, 2025, doi: 10.1109/InCACCT65424.2025.11011351.

[3] Chandan, R. R., Ramaraj, S., Sarishma, & Devi, P. P., et al., *"A Novel Machine Learning-Based Early Warning Detection System for Business Customer Churn,"* in Proc. 5th International Conference on Recent Trends in Computer Science and Technology (ICRTCST), IEEE, 2024, doi: 10.1109/ICRTCST61793.2024.10578489.

[4] Li, Y., & Yan, K., *"Prediction of Bank Credit Customers Churn Based on Machine Learning and Interpretability Analysis,"* Data Science in Finance and Economics, vol. 5, no. 1, pp. 25–42, 2025, doi: 10.3934/DSFE.2025002.